YERMAN J. MEREL G.

Project Engineer | Mechanical Design Engineer | Field Engineer

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PROFESSIONAL SUMMARY

Mechanical Engineer with hands-on experience on two international tunnelling megaprojects — Panama Metro Line 3 and the Hudson River Project (USA) — before completing a B.Sc. degree. Specialized in project engineering support, SOP development, mechanical systems validation, and piping design (P&IDs / isometrics) to ASME B31 standards. Proven track record of delivering corrective engineering solutions under demanding field conditions, reducing execution errors by 20%, and improving audit readiness through regulatory traceability. Seeking to leverage tunnelling and heavy civil infrastructure expertise within a global engineering or EPC organization.

CORE COMPETENCIES

Technical: Field Engineering & Technical Coordination • Structural Validation & FEA (ANSYS) • Industrial Piping Design (P&IDs, Isometrics, ASME B31) • Commissioning & Shutdown Maintenance • Mechanical Systems Validation • CAD Design (AutoCAD, Inventor)

Project & Management: SOP Development & Standardization • Compliance, Traceability & Risk Mitigation • Cross-functional Coordination • Cost Analysis & Resource Allocation

Software: ANSYS (FEA simulations) • MATLAB (numerical modeling) • Python (data automation & reporting) • SAP (PM/MM modules) • AutoCAD • Inventor • MS Office (advanced Excel, Word, PowerPoint)

PROFESSIONAL EXPERIENCE

Co-Founder & Director of Systems Engineering | MONCAN

Panamá City, Panamá • Aug 2025 – Present

Engineering consultancy focused on industrial process optimization and digital workflow design for SMEs.

- Designed the company's operational architecture using systems engineering principles, aligning engineering execution with business strategy.
- Developed digital workflows improving traceability and control of project-related information from client intake to final engineering delivery.
- Built cost-analysis and resource allocation models to support data-driven decision-making and improve operational efficiency.
- Structured scalable internal processes to reduce operational friction and ensure consistent execution across client engagements.

Project Engineer | Herrenknecht Tunnelling Services Panama Corp.

Panamá City, Panamá • Jan 2025 – Jul 2025

Global tunnelling OEM. Supported engineering operations across international TBM projects including the Hudson River Tunnel (USA).

- Developed and standardized a global SOP for critical mechanical maintenance operations on TBM systems covering 3+ active international project sites, improving execution repeatability and reducing non-conformance events.
- Performed structural validation calculations and functional analysis of mechanical assemblies — including main drive components — ensuring safe operation under high-demand industrial conditions; findings contributed to reducing equipment downtime.
- Ensured regulatory compliance and documentary traceability across project milestones, improving audit readiness and reducing contractual risk exposure by consolidating a previously fragmented documentation system.

- Supported engineering reporting and technical coordination between field operations and project stakeholders across USA and international teams.

Key projects: Hudson River Tunnel Project (USA) • Railway Tunnel Project (Perú)

Field Engineer – TBM & Surface Facilities (Intern) | China Railway Tunnel Group Co., Ltd.

Panamá City, Panamá • Aug 2024 – Jan 2025

Contributed to the active tunnelling operations of Panama Metro Line 3, the country's largest infrastructure project.

- Designed corrective engineering solutions for mechanical and piping systems under unstable terrain conditions, eliminating severe operational incidents and improving continuity of TBM advance.
- Developed standardized technical documentation (method statements, inspection checklists, work instructions) adopted as reference material by QC and Engineering teams, reducing execution errors by 20%.
- Produced P&IDs and isometric drawings for high-pressure fluid systems in compliance with ASME B31 standards.
- Supported pressure testing activities, mechanical assembly verification, and system inspections to ensure reliability and safety of critical TBM subsystems.

Key project: Panama Metro Line 3 – Tunnel Section

EDUCATION

B.Sc. in Mechanical Engineering • Technological University of Panama

Panamá City, Panamá • Jan 2018 – Aug 2024

Relevant coursework: Finite Element Analysis (FEA), Computational Modeling, Numerical Methods, Control Systems, Fluid Mechanics, Thermodynamics.

CERTIFICATIONS & TRAINING

- Fluid Technology, Hydraulics & Lubrication – Herrenknecht AG (Jun 2025)
- Design of Thermofluidic Systems – Technological University of Panama (Aug 2023)

In Progress / Planned:

- CAPM – Project Management Professional Certification (Target: Q3 2026)
- OSHA 30-Hour General Industry Safety Certification (Target: Q2 2026)

TECHNICAL PROJECTS & RESEARCH

Resonant Wind Turbine Mathematical Model — *Finalist, National Scientific Initiation Conference (2023)*

- Developed a MATLAB-based mathematical model to analyze resonance behavior in wind turbine structures under dynamic loading.
- Presented results and methodology at a national-level scientific engineering conference, competing against university research teams.

LANGUAGES

Spanish (Native) • **English** (C1 – Full Professional Proficiency) • **German** (A2 – actively developing; strong asset given prior work with Herrenknecht AG)

PROFESSIONAL AFFILIATIONS

- Member, Junta Técnica de Ingenieros y Arquitectos de Panamá (JTIA) – (ID: 2025-016-053)